

NAME: AYOBAMI REBECCA AYOMIDE

MATRIC NO: 241202036

DEPARTMENT: CYBER SECURITY

FACULTY: COMPUTING

COURSE CODE: COS 201

COURSE TITLE: COMPUTER PROGRAMING II(PYTHON)

Assignment

1. Using PYTHON programming language, develop an interactive and well-designed screen MATHEMATICAL CALCULATOR (MC) program containing components such as +, -, /, *, ^, %, C (Clear), OFF etc. (NB: It's a simple calculator not a complex scientific one.)

Answer

```
print("=====")
print("SIMPLE MATHEMATICAL CALCULATOR")
print("=====")
```

```
while True:
```

```
    print("1. Addition (+)")
    print("2. Subtraction (-)")
    print("3. Multiplication (*)")
    print("4. Division (/)")
    print("5. Modulus (%)")
    print("6. Power (^)")
    print("7. Clear ©")
    print("8. OFF")
```

```
    choice = input("Enter your choice (1-8): ")
```

```
    match choice:
```

```
        case "1":
```

```
            print("You selected Addition")
```

```
            num1 = int(input("Enter first number: "))
```

```
            num2 = int(input("Enter second number: "))
```

```
addition = num1 + num2  
print("Addition =", addition)
```

case "2":

```
Print("You selected Subtraction")  
num1 = int(input("Enter first number: "))  
num2 = int(input("Enter second number: "))  
subtraction = num1 - num2  
print("Subtraction =", subtraction)
```

case "3":

```
print("You selected Multiplication")  
num1 = int(input("Enter first number: "))  
num2 = int(input("Enter second number: "))  
multiplication = num1 * num2  
print("Multiplication =", multiplication)
```

case "4":

```
print("You selected Division")  
num1 = int(input("Enter first number: "))  
num2 = int(input("Enter second number: "))  
if num2 == 0:  
    print("Division by zero is not allowed.")
```

else:

```
    division = num1 / num2  
    print("Division =", division)
```

case "5":

```
print("You selected Modulus")  
num1 = int(input("Enter first number: "))
```

```
num2 = int(input("Enter second number: "))  
modulus = num1 % num2  
print("Modulus =", modulus)  
case "6":  
    print("You selected Power")  
    num1 = int(input("Enter first number: "))  
    num2 = int(input("Enter second number: "))  
    power = num1 ** num2  
    print("Power =", power)  
case "7":  
    print("Calculator Cleared!")  
case "8":  
    print("Calculator OFF")  
    print("Thank you for using the calculator.")  
    break  
case _:  
    print("Invalid choice. Please try again.")
```

Output

SIMPLE MATHEMATICAL CALCULATOR

1. Addition (+)
2. Subtraction (-)
3. Multiplication (*)
4. Division (/)
5. Modulus (%)
6. Power (^)
7. Clear ©

8. OFF

Enter your choice (1-8): 4

You selected Division

Enter first number: 45

Enter second number: 5

Division = 9.0

1. Addition (+)

2. Subtraction (-)

3. Multiplication (*)

4. Division (/)

5. Modulus (%)

6. Power (^)

7. Clear ©

8. OFF

Enter your choice (1-8): 5

You selected Modulus

Enter first number: 13

Enter second number: 4

Modulus = 1

1. Addition (+)

2. Subtraction (-)

3. Multiplication (*)

4. Division (/)

5. Modulus (%)

6. Power (^)

7. Clear ©

8. OFF

Enter your choice (1-8): 8

Calculator OFF

Thank you for using the calculator.

[Program finished]

- 2.** CGPA is not something that should constrain you to visiting AVERS at all time.
Using the powerful provision of selection control statements in PYTHON,
develop a PERSONAL POCKET CGPA Calculator (PPC).

Answer

```
Print("=====")
```

```
Print("PERSONAL POCKET CGPA CALCULATOR")
```

```
Print("=====")
```

```
total_grade_point = float(input("Enter Total Grade Point: "))
```

```
total_credit_unit = float(input("Enter Total Credit Unit: "))
```

```
cgpa = total_grade_point / total_credit_unit
```

```
print("CGPA =", cgpa)
```

```
if cgpa >= 4.50 and cgpa <= 5.00:
```

```
    print("Class of Degree: First Class")
```

```
elif cgpa >= 3.50 and cgpa <= 4.49:
```

```
    print("Class of Degree: Second Class Upper")
```

```
elif cgpa >= 2.40 and cgpa <= 3.49:
```

```
    print("Class of Degree: Second Class Lower")
```

```
elif cgpa >= 1.50 and cgpa <= 2.39:
```

```
        print("Class of Degree: Third Class")
    elif cgpa >= 1.00 and cgpa <= 1.49:
        print("Class of Degree: Pass")
    else:
        print("Class of Degree: Fail")
```

Output

PERSONAL POCKET CGPA CALCULATOR

Enter Total Grade Point: 69.56

Enter Total Credit Unit: 21

CGPA = 3.3123809523809524

Class of Degree: Second Class Lower

[Program finished]

The programs were successfully developed using Python, tested, and uploaded to GitHub.